Computational Coupling — Project Information

Contents

1. The question, and why it was open . . . 1
2. What is new here . . . 2
3. Data . . . 4
4. Method . . . 5
5. Every experiment we ran . . . 6
6. Results . . . 6
7. What this does not claim . . . 8
8. Where our own predictions failed . . . 8
9. Repository layout . . . 9
10. Reproducing everything . . . 9
11. Testing . . . 10
12. The audit, and the pattern in it . . . 10
13. Relationship to v4 . . . 11
14. Open work . . . 11

Four choices decide whether a connectome beats its rewiring.

Computational coupling is the extent to which what a network computes is attributable to how it is wired. It is never measured directly — only as the gap between a network's score and the scores of surrogate networks, under some unit model, at some operating point, through some readout. This project shows that each of those four choices can flip the sign of that gap on identical data, so a coupling reported as a single signed number is under-determined by its own methods section.

Authors — Ashok Raj Pasala · Snigdha Gorai · under the guidance of Jeevana Jyothi P.

Affiliation — School of Computer Science and Engineering, VIT-AP University, Amaravati 522237, Andhra Pradesh, India

Version — 5.0, supersedes and retracts v4 · **Licence** — MIT

Manuscript — `paper/computational_coupling_v5.pdf` (27 pages, two-column) · **Record of withdrawals** — `AUDIT.md`

This document is the single reference for the project: what it asks, how it is built, what it found, what it does not claim, and how to reproduce every number in it. The manuscript is the scientific statement; this is the operational one.

---

1. The question, and why it was open

Connectomes contain statistically overrepresented subgraphs. A large literature holds that these *motifs* are what make biological wiring computationally effective; a neuromorphic literature builds architectures on that premise.

Testing the claim needs a control that differs from a connectome **in its motif structure and in nothing else**. No such control existed. Rewiring a connectome perturbs its motif census together with every statistic that co-varies with it — clustering, spectrum, higher-order structure — so a performance difference against a rewired null cannot be attributed to motifs in particular.

Sampling the ensemble that would settle it is hard: it requires moving through the space of degree-constrained graphs under a constraint on a *global* subgraph count, with the census recomputed at every proposal.

Three further confounds recur in this literature. All three appear in our own v4 work, and each is a failure that produces no error and no warning:

1. **A "Dale-compliant" null imposes no constraint.** When synapses inherit their presynaptic neuron's valence, Dale's principle is automatically preserved by any double-edge swap. Such a null is the standard Maslov–Sneppen ensemble and cannot test whether the *placement* of inhibition carries information.
2. **An all-to-all reservoir does not use its wiring.** With every neuron driven and every neuron decoded, the task is solved by the input projection and the leak. We measure 59.6% for a **zeroed** recurrent matrix against 59.6% for the connectome. Under the sensory/motor-disjoint protocol used throughout this work the same control falls to 4.9% — chance — while the connectome reaches 35.2%. That gap is the diagnostic that the benchmark is sensitive to recurrent structure at all.
3. **Matched spectral radius is not a matched operating point.** Erdős–Rényi surrogates settle at 1.77× the firing rate of the connectomes they are matched to, and rate predicts accuracy ($r = 0.38$). A matched-gain comparison confounds topology with activity level, in the direction that flatters the unstructured control.

2. What is new here

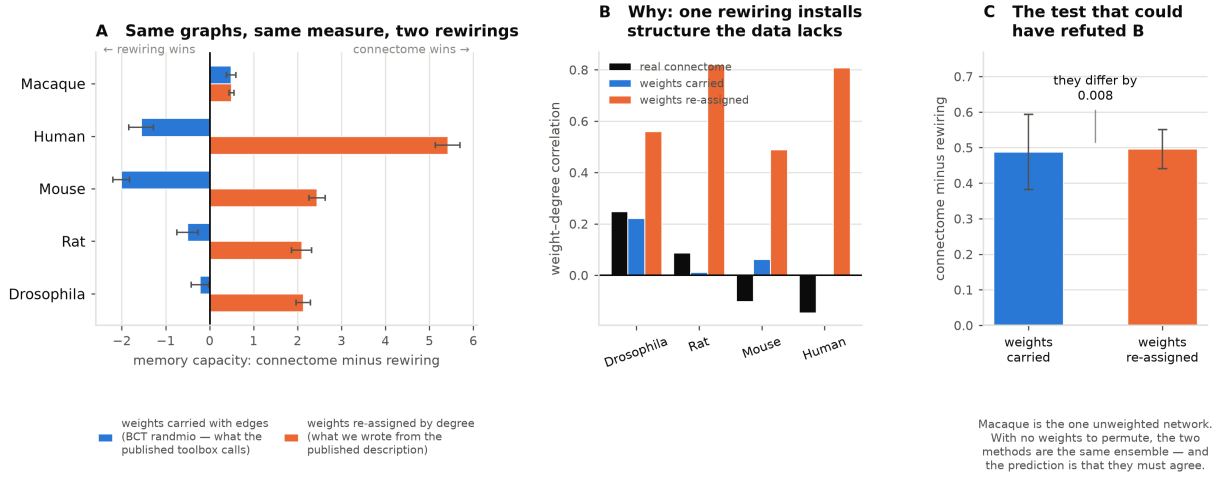
1. The connectome advantage does not survive a firing threshold — `experiments/ladder/run_unit_model_compari` `experiments/ladder/run_null_mechanism.py`

The reservoir literature reports that empirical connectomes beat degree-preserving rewirings on memory capacity, and that result is increasingly invoked as motivation for neuromorphic architectures. We ran identical graphs, identical inputs, delays and readout through two reservoirs differing **only** in their units, against **two** rewiring ensembles that both answer to "degree-preserving randomisation", on the five regional connectomes distributed with the original toolbox — 2,700 evaluations, 5 networks × 10 paired instances × 9 gains × 2 unit models × (connectome + 2 nulls).

The null choice changes the answer. "Degree-preserving randomisation" names at least two ensembles: one where a weight travels with its edge (BCT `randmio_*_connected`, what `conn2res` calls), one where the weight multiset is re-assigned to the rewired graph by endpoint-degree rank. We wrote the second believing it was the first.

null ensemble	connectome beats it at ρ=1, graded units	median difference
BCT `randmio` (the published one)	**1 of 5** leads, **3 of 5** trail	−0.51
rank-reassigned (ours, mistakenly)	**5 of 5** lead, all significant	+2.13

Of the 10 (network, unit model) cells, the two nulls give opposite **signs** in 6 and opposite **significant verdicts** in 3. Median disagreement 0.73 capacity units, max 6.98.



(A) The result. (B) The cause: one rewiring installs a weight-degree association the real graphs do not have. (C) The test that could have refuted (B) — on the one network with no weight variation, the two procedures must agree, and they do, to 0.008.

Why, and the prediction that could have refuted it. Rank-reassignment does not only randomise — it *installs* a weight–degree association the data lacks, shifting that correlation by a median of **+0.66** across the four weighted networks against **+0.06** for weight-carrying rewiring. So on a network with **no weight variation** the two must coincide exactly. Macaque is binary. There they shift the association by 0.000 and their measured advantages differ by 0.008 capacity units — the one cell where the choice provably cannot matter is the one cell where it does not.

This is **not** a refutation of the published result: our harness also differs in its input projection, delay set and ridge solver. It shows that the null choice *alone* moves the result by more than the effect being reported, which is a sufficient explanation for a reproduction failure without anything else being wrong.

units	connectome beats `randmio` at ρ=1	across the gain axis	silent units
graded `tanh`	**1 of 5** leads, 3 trail	29% of cells	0.6%
spiking LIF	**0 of 5**	9% of cells	60.7%

One mechanism changed sign when the null was corrected. We predicted a threshold would silence a heavy-tailed graph's low-degree neurons differentially. Silencing is large (60.7% vs 0.6%). Against the rank-reassigned null the silencing gap did *not* predict the capacity gap and we reported the prediction as failed; against `randmio` it **does** — $r = -0.58$, $p = 2.5\times10^{-5}$ over 45 (network, gain) cells, in the predicted direction. We report both, because a wrong control manufactured a clean positive result in one place and destroyed a real one in another, from the same code and the same graphs. Thresholding also compresses capacity for every topology alike, by a factor of 4.9, and that part is not differential at all.

A floor, declared. The human network's spiking capacities are 0.049 vs 0.019 — both effectively zero, so it carries nothing either way. The claim rests on the other four, which retain capacities of 1.0–1.9.

2. The contrast also changes sign along the gain axis — `experiments/ladder/run_gain_sweep_ladder.py`

3. A motif-preserving null ensemble — `src/nulls/swap_chain.py`

Degree-, valence- and census-matched surrogates, sampled by an annealed Metropolis chain over exact, incrementally updated motif counts.

We do not claim the idea. Motif-constrained ensembles are standard in statistical network science: Milo et al. proposed a simulated-annealing null preserving lower-order subgraph counts; exponential random graph models condition on triad counts by design (Frank & Strauss 1986; Robins et al. 2007); the dk-series is a nested hierarchy of nulls constraining progressively higher-order structure (Mahadevan et al. 2006; Orsini et al. 2015); and Fischer et al. (*PRL* 2015) sample motif-constrained ensembles at fixed degree sequence by Wang–Landau multicanonical sampling, which unlike annealing comes with a sampling guarantee. What we contribute is an implementation for the **sign-attributed** census at connectome scale — exact coherent/incoherent FFL counts under Dale's principle, O(d) incremental updates, and a move set that survives near-total reciprocity — and the use of that ensemble as a **control rung** rather than as an enrichment null. Where a formally exact sampler is available it should be preferred.

Two things about the construction matter, because the obvious implementation does not work:

- *Where the chain starts decides which property is guaranteed.* Starting from a fully rewired graph guarantees randomisation and leaves convergence to be shown; starting from the connectome under a census constraint guarantees convergence and leaves randomisation to be shown. We use the empirical start and demonstrate randomisation by edge overlap, because the random start fails outright on networks whose census is hard to rebuild by chance.
- *Ordinary swaps destroy reciprocal pairs faster than they create them.* On a near-symmetric network the reciprocal-pair count drifts down however stiff the penalty. We add a move that swaps two reciprocal **pairs** as units, preserving both degrees and reciprocity.

4. A five-rung structural control ladder — `src/nulls/ladder.py`

Rung	Preserves	Isolated by comparing with the rung below
L0	N, edge count, E:I ratio	—
L1	+ joint in/out degree sequence	degree heterogeneity
L2	+ valence–topology coupling	where inhibition sits
L3	+ 3-node motif census	motif structure
L4	the empirical connectome	residual higher-order wiring

5. Two controls that are usually missing — `experiments/ladder/`

The zero-recurrence control, and the ladder re-run with every network individually recalibrated to a common firing rate.

6. A gap-free perturbation bound — `src/theory/transfer_perturbation.py`

Davis–Kahan is inapplicable to measured connectomes: the smallest gap we measure is 4.4×10^{-2}, admitting perturbations of norm at most 2.2×10^{-2} — less than removing one synapse. We bound the finite-horizon input–output map instead, which needs no gap, and extract the **operational alignment** γ of a perturbation.

7. An honest hardware report — `experiments/hardware/run_open_synthesis.py`

Real logic synthesis with Yosys + ABC onto the Nangate45 open cell library, reporting only what such a flow establishes.

3. Data

Network	Source	N	|E|	Valence
C. elegans whole nervous system	Cook et al. 2019	300	5,044	yes
Drosophila central complex	FlyWire v630	400	7,856	yes
Drosophila antennal lobe	FlyWire v630	400	16,468	yes
Mouse V1 L2/3	MICrONS Pinky100	334	1,734	**no**
Human, regional	HCP consensus tractography	400	6,092	**no**

Four public releases spanning three phyla, yielding five networks. Two carry synaptic valence and two do not, and that constrains which questions each can answer:

- **Mouse.** The release annotates 34 inhibitory somata, but every one is **disconnected** in the accompanying synapse table — all 1,736 synapses are excitatory→excitatory. Treated as topology-only. Reporting coherent feedforward loops here would be reporting plain feedforward loops under another name.
- **Human HCP.** Diffusion tractography carries no synaptic valence. Treated as topology-only, and labelled a *regional* network rather than a neuronal connectome.
- **Human microscale is not analysed.** The H01 release provides cell and synapse tables, but the connectivity export needed to build a wiring diagram is not in the distribution we worked from. A census of a synapse table is not a connectome analysis. (v4 claimed this tier; see §12.)

Task data. Spiking Heidelberg Digits (Cramer et al. 2022): 20 classes, 700 cochlear channels, 12 speakers, canonical speaker-held-out split. Binned to 100 time steps over 1 s, channels pooled to 140 bands. A real spike-based benchmark, replacing the synthetic tasks of v4 which saturated near 100% for every architecture.

Subcircuit extraction. Where a release exceeds the common reservoir size, we take an induced subgraph by breadth-first expansion from an above-median-degree seed, then drop isolates. Growing from a hub could inflate degree heterogeneity — the property the largest effect is attributed to — so we measured it: the ratio of subgraph to full-release degree CV is at most **1.00** across all five networks, as low as **0.79** among the three genuinely subsampled. Extraction biases the degree result *against* itself, if at all.

4. Method

Exact sign-attributed motif census

Under Dale's principle the sign product around a feedforward loop ($u\to v$, $v\to w$, $u\to w$) is $\sigma_u \cdot \sigma_v \cdot \sigma_u = \sigma_v$.

> **A feedforward loop is coherent if and only if its *intermediate* neuron is excitatory.** Coherence is a property of one node, not of a product over three edges.

This makes the sign-attributed census exactly as cheap as the topological one. All counts are exact; none are estimated. Single-edge insertions and deletions admit exact $O(d)$ updates of every 3-node count, which is what makes the annealed null tractable.

The annealed chain

Energy is the squared deviation of the running census from a target, **measured in units of the unconstrained-ensemble standard deviation** — over reciprocal pairs, mutually inhibitory pairs, FFL, cFFL, iFFL and 3-cycles, across $120 \times |\mathrm{E}|$ proposals. Bi-fan is deliberately excluded and reported post hoc as a **held-out statistic**.

The normalisation lives in exactly one place (`census_scale`). An earlier version restated it at ten call sites with an extra floor of 2% of the empirical count, which made the constraint roughly five times slacker than the prose described and left the chain unable to hold the census on human tractography. See `AUDIT.md` §D2.

Evaluation protocol

Fixed reservoir, trained ridge readout only. Discrete-time LIF units, leak 0.90, threshold 1. Held fixed across every network in a comparison: node count, edge count, spectral radius ($\rho = 1.30$) and input projection.

Sensory/motor disjoint. The input reaches 15% of neurons; the readout sees a **disjoint** 35%. Task-relevant signal must traverse the recurrent graph to be decodable. The zero-recurrence control then falls to chance, which is the diagnostic that the benchmark is sensitive to recurrent structure at all.

Statistics. Welch's t with Holm correction across the full 20-contrast family, percentile-bootstrap CIs, Hedges' g. 15 surrogates per rung. Rung L4 is one graph, so its replicates carry one variance component where L0–L3 carry two; a contrast involving L4 supports inference about *this* connectome rather than connectomes as a class.

Power. 80% power to resolve $d = 1.06$ uncorrected, $d = \mathbf{1.54}$ at the Holm worst case. Stated so a small measured effect can be told from an underpowered design.

5. Every experiment we ran

15 of 15 experiments have results on disk. Sizes are counted from each experiment's own output file when this document is built, so they cannot drift from what was actually run. The unit differs by row: a "run" is one network at one rung trained to convergence, an "evaluation" one network at one operating point probed for one metric, so the column does not sum to anything meaningful.

#	Experiment	What it asks
1	**Motif enrichment** `experiments/motifs/run_motif_significance.py`	Are the classic su
2	**Motif-preserving null ensemble** `experiments/ladder/run_null_validation.py`	Validation of the
3	**Sampler robustness** `experiments/ladder/run_null_robustness.py`	The same ensemb
4	**Protocol controls** `experiments/ladder/run_protocol_controls.py`	What does the be
5	**Structural control ladder** `experiments/ladder/run_structural_ladder.py`	L0 density to L4
6	**The ladder at matched firing rate** `experiments/ladder/run_rate_matched_ladder.py`	The same decomp
7	**Extraction check** `experiments/ladder/run_extraction_check.py`	Does taking a siz
8	**Two null ensembles, two unit models** `experiments/ladder/run_unit_model_comparison.py`	The headline exp
9	**Why the two nulls differ** `experiments/ladder/run_null_mechanism.py`	Structural descrip
10	**Spiking tuning control** `experiments/ladder/run_spiking_tuning_control.py`	Input gain x thre
11	**Gain sweep of the ladder** `experiments/ladder/run_gain_sweep_ladder.py`	The whole decom
12	**Memory-capacity sweep** `experiments/ladder/run_memory_capacity_sweep.py`	Locating the pub
13	**Intrinsic dynamics** `experiments/ladder/run_dynamics_ladder.py`	The same decomp
14	**Pruning frontier** `experiments/degeneracy/run_pruning_frontier.py`	Random, hub-tar
15	**Open-PDK synthesis** `experiments/hardware/run_open_synthesis.py`	Real logic synthes

Three of these exist only because they could have refuted us, and two of them partly did: the **spiking tuning control** (#10) searched for the configuration that would overturn our own spiking result and found enough to force the claim down from "absent" to "sporadic and inconsistent in sign"; the **pruning frontier** (#14) tested a scaling prediction of our own perturbation bound and falsified it, leaving only the ordering; and the **extraction check** (#7) tested whether our own subgraph sampling was producing the degree result. We regard running them before submission rather than after as the difference between a finding and a correction.

6. Results

Motif enrichment, recomputed

500 degree-preserving surrogates, exact counts, resolution floor stated. Enrichment is real but **not conserved**:

Outcome	Count
Significantly enriched	23 / 29
Significantly **depleted**	3 / 29
Indistinguishable from null	3 / 29

Coherent feedforward loops are **depleted** in the *Drosophila* central complex ($Z = -10.9$). Mouse V1 sits at its null throughout (no motif above $Z = +2.7$). The normalised significance profiles of the two fly circuits agree with each other no better than either agrees with the worm.

Null ensemble validation

Network	Max rel. error	Converged	Overlap L3	Overlap L2
C. elegans	0.25%	100%	13.9%	11.6%
Drosophila CX	0.20%	100%	20.5%	19.9%
Drosophila AL	0.11%	100%	43.9%	40.0%
Mouse V1	0.85%	100%	10.6%	10.7%
Human HCP	0.50%	100%	11.4%	7.8%

The chain hits its target, randomises to near the unconstrained level, and does **not** reproduce the held-out bi-fan count — nor displace it in a consistent direction (overshoot in the worm, undershoot in the fly CX), which is what distinguishes L3 from a partial reconstruction of the connectome.

The constraint is binding. L2 and L3 are separated by tens to hundreds of null standard deviations in the constrained coordinates — except in mouse V1, where the census sits near its null by construction and the contrast is correspondingly uninformative.

Not a sampler artefact. Over an eightfold range of chain lengths accuracy moves by a median of 1.04 pp, non-monotonically in 4 of 5 networks. The opposite construction (random start, annealing towards the census) converges on 4 of 5 and agrees to within 1.07 pp where it does.

The structural decomposition

Every structural property that measurably moves accuracy moves it down.

Property added	Significant	Median	Largest	Hedges' g
Degree sequence	3 / 5	**−4.65 pp**	−8.02 pp	−2.38
Motif census	2 / 5	**−3.50 pp**	−5.94 pp	−2.06
Valence placement	1 / 5	−0.82 pp	−3.16 pp	−1.14
Residual wiring	2 / 5	+1.06 pp	−7.60 pp	−2.04

The motif contrast is **not** negligible — it is comparable in size to the degree sequence — but its sign is negative. Imposing a connectome's motif structure on a degree- and valence-matched surrogate **costs** accuracy on this task. That is the opposite sign to the one the motif hypothesis predicts.

At a matched operating point

Recalibrating each network to a common mean firing rate of 0.07 does **not** rescue the connectomes: the Erdős–Rényi rung still leads by 9.00–13.95 pp, marginally more than at matched gain (7.96–11.74 pp). 67–91% of the decline is spent at the step that imposes the degree sequence.

27 of 250 calibrations never reached the target, 25 of them in the *Drosophila* central complex, whose edges are 66% inhibitory — raising its gain raises its inhibition, so its attainable rate is capped by its own valence organisation. Those cells are excluded from every matched-rate statistic and marked in the table rather than averaged in.

Dynamical probes

Task-independent measures (memory capacity, kernel rank, generalisation rank) reproduce the ordering: the Erdős–Rényi rung has higher memory capacity in **5 of 5** networks (median 1.41×) and higher kernel rank in 5 of 5 (1.21×). The connectomes are not trading task accuracy for capability the benchmark fails to reward.

A disagreement we do not resolve. The probes and the task metric agree on direction and on the degree sequence carrying the largest share, but disagree on the motif rung: comparable to degree on accuracy, smallest in the ladder on the probes. The manuscript states the divergence and offers a conjecture labelled as one.

Degeneracy

Tolerated pruning (largest removal retaining 90% of unpruned accuracy):

Direction	Tolerated	Median γ, relative to random
Uniformly random	**83%** (lower bound)	1.0×
Degree-targeted	73%	2.6×
Adversarial (first-order)	64%	3.0×

All three strategies **improve** accuracy at low fractions, peaking around 40–60% removal. These reservoirs are over-connected for the task, consistent with degree heterogeneity being costly here.

Hardware

Yosys + ABC onto Nangate45, typical corner:

Module	Cells	Flip-flops	Cell area (μm^2)
`dendritic_lif_core`	4,401	70	5,301
`synaptic_accumulator`	19,649	8,192	63,361
`aer_crossbar_router`	39	17	124
`cen_accelerator_top`	590,116	25,644	749,263

No die area, no power, no energy per operation, no clock frequency. The flow performs no placement, routing, clock-tree synthesis or parasitic extraction, and the toolchain ships no static timing analyser.

7. What this does not claim

- **Not** that connectome wiring is bad for computation. It is a poor fixed reservoir for a 20-class classification with a linear readout — one task, one regime, and not the objective any nervous system is under selection for. The uniform negative sign is informative about the probe: a linear decoder rewards distributed, high-dimensional state, and connectome structure concentrates and correlates activity.
- **Not** that motif structure is functionally irrelevant. The design resolves $d = 1.54$ after correction; non-significant contrasts do not exclude moderate effects.
- **Not** anything about **learnability**. Motif structure has been argued to act as an inductive bias, which is a claim about learning trajectories, not about the function computed by a frozen network. Testing it needs a trained ladder (§13).
- **Not** any silicon figure beyond gate count, flip-flop count and pre-layout cell area.
- **Not** a microscale human result.

8. Where our own predictions failed

Recorded because narrowing a claim quietly after seeing the data is the behaviour this project exists to correct.

The $1/\gamma$ scaling. Proposition 2 predicts tolerance scales as $1/\gamma$. The **ordering** holds — γ ranks the directions correctly in 3 of 5 networks. The **scaling** does not: the product $\gamma \cdot \|\mathrm{E}\|\max$ should be constant across directions and varies by a factor of 2.2. γ identifies which direction is dangerous and does not predict by how much.

γ is not comparable between networks. It is a realised fraction of a worst-case constant depending on $\|\mathrm{A}\|_2{}^T$, and its median spans ~7×10^5 across our five networks. Pooled γ medians — which our first analysis reported, and our first figure plotted — are not a quantity at all.

The rate-matching expectation. We introduced the matched-rate ladder expecting it to explain the Erdős–Rényi advantage. It does not; the gap widens slightly. The resulting claim is stronger and less comfortable than the one we set out to make.

The operating-point expectation, twice. A one-instance probe suggested the connectome-versus-null memory-capacity contrast turned positive exactly at criticality and negative on both sides — a clean crossing that would have reproduced and located the published result. The paper was rewritten around it. At ten paired instances the pattern was scatter, not a crossing: 0 of 5 significant. The obvious next move, reporting a failed replication, would have been wrong too — the design is underpowered there by roughly a factor of five. The honest report is a non-detection with its power stated. See `AUDIT.md` §E4.

The motif artefact expectation. The human motif contrast was significant under a sampler that never converged on that network. We expected it to vanish once the sampler was fixed, and recorded that expectation before the re-run. It did not vanish — it persists at $g = -2.06$, joined by *C. elegans* at -1.46. Had it vanished, this paper would have reported a clean null on motifs; instead it reports a real effect pointing the opposite way to the hypothesis under test.

9. Repository layout

`src/motifs/fast_census.py`	exact sign-attributed census, O(d) incremental updates
`src/nulls/swap_chain.py`	degree-preserving chains + the annealed motif-preserving null
`src/nulls/ladder.py`	the five-rung structural control ladder
`src/theory/transfer_perturbation.py`	gap-free bound, operational alignment, edge rankings
`src/eval/reservoir.py`	reservoir protocol (sensory/motor disjoint)
`src/eval/dynamics.py`	memory capacity, kernel & generalisation rank
`src/eval/trained_snn.py`	surrogate-gradient training of masked recurrent SNNs
`src/tasks/shd.py`	Spiking Heidelberg Digits loader
`src/analysis/`	figure, table and LaTeX-macro generation
`src/analysis/rebuild_paper.sh`	the whole path from results/ to the compiled PDF
`experiments/`	experiment drivers, one per result
`hardware/rtl/`	Verilog sources
`paper/`	the current manuscript and its generated inputs
`pitch/`	the current executive summary
`AUDIT.md`	record of withdrawn claims and of defects in the replacement
`info.md`	this document
`paper/superseded_v4/`	the retracted v4 manuscript, figures and pitch dossier
`results/superseded_v4/`	the result files those claims were drawn from

Other packages under `src/` (`architecture/`, `benchmarks/`, `learning/`, `remodeling/`, `simulation/`) and the matching directories under `experiments/` belong to v4 and are not exercised by any current result. They are kept because `AUDIT.md` cites them as the origin of specific withdrawn numbers; deleting them would make the retraction harder to check.

10. Reproducing everything

```
python -m venv .venv && .venv/bin/pip install -e ".[dev]"

# Task data (~350 MB)
mkdir -p data/raw/shd && cd data/raw/shd
curl -L -O https://zenkelab.org/datasets/shd_train.h5.gz
curl -L -O https://zenkelab.org/datasets/shd_test.h5.gz && gunzip *.gz && cd -
```

Then the experiments. This list is generated from the same source as the table in §5, so it cannot fall behind what the repository actually contains:

```
python experiments/motifs/run_motif_significance.py      # Motif enrichment
python experiments/ladder/run_null_validation.py         # Motif-preserving null ensemble
python experiments/ladder/run_null_robustness.py         # Sampler robustness
python experiments/ladder/run_protocol_controls.py       # Protocol controls
python experiments/ladder/run_structural_ladder.py       # Structural control ladder
python experiments/ladder/run_rate_matched_ladder.py     # The ladder at matched firing
↪  rate
python experiments/ladder/run_extraction_check.py        # Extraction check
python experiments/ladder/run_unit_model_comparison.py   # Two null ensembles, two unit
↪  models
python experiments/ladder/run_null_mechanism.py          # Why the two nulls differ
python experiments/ladder/run_spiking_tuning_control.py  # Spiking tuning control
python experiments/ladder/run_gain_sweep_ladder.py       # Gain sweep of the ladder
python experiments/ladder/run_memory_capacity_sweep.py   # Memory-capacity sweep
python experiments/ladder/run_dynamics_ladder.py         # Intrinsic dynamics
python experiments/degeneracy/run_pruning_frontier.py    # Pruning frontier
python experiments/hardware/run_open_synthesis.py        # Open-PDK synthesis
```

Then the statistics, figures, tables, macros and three LaTeX passes:

```
./src/analysis/rebuild_paper.sh
```

`rebuild_paper.sh` reports unresolved numbers, undefined references, LaTeX errors and overfull boxes at the end. A clean exit status is not evidence of anything — `pdflatex` exits 0 on a hard error by design, and a duplicate `\newcommand` builds a complete PDF while silently keeping whichever definition came first.

Every number in the manuscript is emitted into `paper/generated_numbers.tex`; none is transcribed by hand, and a missing result appears as a visible `??` rather than as a stale value. Protocol constants are read from the experiment's own argument parser and from the task loader's signature, not restated.

11. Testing

```
.venv/bin/pytest          # 108 tests
```

Covers the census against brute-force enumeration, exactness of incremental updates, degree and valence preservation in every null, convergence and randomisation of the annealed chain, the reservoir protocol controls, and the perturbation bounds.

Every defect in `AUDIT.md` §D is pinned by a regression test that was verified to fail against the old code before being kept. A test that passes against both versions pins nothing, and two of ours originally did.

12. The audit, and the pattern in it

`AUDIT.md` documents 34 entries: **15** in the retracted v4 manuscript (**A**–**C**), **12** found in the *replacement* analysis while building it (**D**–**D12**) — our own, on the same terms — and **7** where a correction changed a conclusion (**E**–**E7**), including two that reversed when we discovered our null ensemble was not the published one.

One pattern runs through nearly all of them:

> **Every defect produced a plausible number and no error.** A loader wrapped in `except Exception: pass` returned a zero-edge graph. `argsort` over constant magnitudes returned index order, turning "weakest-first pruning" into whole-neuron ablation. A null model imposed a constraint it did not impose. An energy scale was five times slacker than the prose claimed. A bound was stated in the spectral radius where the operator norm was required — in a paper

arguing that eigenvalue reasoning fails on these networks. A protocol control decoded nothing and scored chance. `pdflatex` reported a duplicate macro and exited 0 with a complete PDF.

None was caught by a result looking wrong. Each was caught by checking a recorded diagnostic, or by reading the code against the prose. That is why the convergence diagnostics, the protocol control and the held-out statistic are reported in the paper rather than kept in the repository.

13. Relationship to v4

This work **supersedes and retracts** the previous manuscript. Withdrawn:

- **Commercial silicon figures that were never measured.** "TSMC 28nm HPC+", a die area, 116.71 pJ/SynOp, 2,857× real-time acceleration, a "physical tapeout". No Cadence tool, no TSMC library and no synthesis output ever existed in the repository; the numbers came from a Python model with those values as literals. The RTL had never been through any synthesis tool, and did not compile.
- **A human microscale tier built on an empty graph.** `load_human_h01_microcircuit()` returned a graph with **zero edges**; the loader wrapped the failure in `except Exception: pass`. The "26.5M synapses" figure is a sum over a column in a cell table.
- **Sign-attributed motifs for datasets with no usable inhibition.** Mouse and human entries where cFFL equals FFL by construction.
- **p-values below the resolution of the null that produced them.** `p < 0.0001` from K = 25 surrogates, where the smallest attainable value is $1/(K+1) \approx 0.038$.
- **An ablation table whose own control row contradicted its conclusion.** The Erdős–Rényi control was *better* than the motif-structured model on the very metric used to argue for motifs.
- **A "455× energy reduction"** comparing a near-silent network against a dense one firing freely, at different accuracies.

The retracted documents are at `paper/superseded_v4/` and their data at `results/superseded_v4/`, labelled. The v4 **executive pitch dossier** — the artefact most likely to be forwarded — carries its own retraction notice at `paper/superseded_v4/pitch/RETRACTED.md`.

14. Open work

The trained ladder. `experiments/ladder/run_trained_ladder.py` exists and is **not** run or reported. It tests motifs as an inductive bias for learning, which the fixed-reservoir design cannot address. Its harness has been corrected — it previously used the all-to-all protocol this paper rejects — and its firing-rate penalty retuned, but it needs a zero-recurrence control that separates from the connectomes before any number from it means anything. One trap is documented: with only a fraction of neurons driven, a firing-rate penalty strong enough for an all-to-all network makes total silence the global optimum, and a collapsed network returns a perfectly flat ladder indistinguishable from the result reported here.

A weighted null. All weights here are binary up to valence. Real synapses have graded strengths correlating with topological position. The annealed chain generalises to weighted targets without modification; the degree-preserving move set does not.

An exact sampler. Annealing is not proven to sample uniformly over the constraint set. We show the conclusion survives an eightfold variation in chain length and does not depend on the starting point, which rules out two specific ways an annealed sampler could manufacture the result — but a formally exact sampler for a census-constrained degree-preserving ensemble remains open.

A better-powered motif contrast. A design powered for $d \approx 0.5$ at this correction would need of order a hundred surrogates per rung.

γ against measured biological perturbation. The prediction is that developmental and activity-dependent rewiring occupies the low-γ regime that makes degeneracy cheap.